

# An Explanatory Analysis of Power Factor Efficiency and Peak Loads in Steel-Plant Operations

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## Introduction

Industrial energy use accounts for a large share of global consumption, making efficiency improvements both vital and cost-effective. This study analyzes one year of 15-minute operational data from a small steel plant in Gwangyang, South Korea, gathered through IoT monitoring. The dataset captures electrical load and power-quality measures such as reactive power and power factor. Using an explanatory approach, we investigate two key issues: high-load intervals and low power-factor periods through binary logistic regression to identify patterns driven by time, load type, and reactive-power behavior. The goal is to provide actionable insights for optimizing operations, improving efficiency, and reducing grid stress.

## Objectives

### I. Factors Contributing to High Load Periods:

A High Load event is defined as any 15-minute interval where energy consumption exceeds the 95th percentile of the dataset, isolating only the highest levels of usage while avoiding rare outliers. This binary classification (High vs. Low load) provides a practical benchmark consistent with industry standards. A logistic regression model will then be developed to estimate the likelihood of high-load occurrences based on predictors such as time of day, day type, and power-quality factors.

### II. Factors Affecting Power Efficiency:

A Power Factor Efficiency event is defined using a 0.90 threshold on the lagging current power factor. Intervals with PF values below 0.90 are classified as inefficient, while those at or above 0.90 are labeled Efficient. This benchmark aligns with common utility standards, as maintaining a PF of 0.9 or higher helps prevent penalties and reduces power losses. A logistic regression model will identify factors such as reactive power, load type, and time that increase the likelihood of inefficiency.

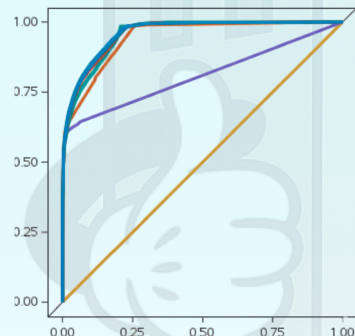
## Methodology

Binary logistic regression models the probability of an event occurring such as a high-load demand or low power-factor period by relating predictor variables to the log-odds of that event.

A stepwise selection procedure was employed to choose only variables that significantly improve model fit ( $\alpha = 0.05$ ). Stepwise regression iteratively adds or removes predictors based on statistical contribution, achieving a parsimonious yet robust model while avoiding redundancy and multicollinearity. This yields interpretable coefficients and odds ratios that quantify how each factor influences the likelihood of a high-load or inefficient interval.

We adopt an explanatory approach because the [original research already addressed prediction](#) using algorithms such as Cubist and SVM. [Our logistic models instead uncover why inefficiency or peaks occur](#), using interpretable odds ratios to guide strategies like load shifting and targeted power-factor correction. The models include six predictors: Load Type, Day of Week, Month, Day, Usage (kWh), and Reactive Power (Leading/Lagging kVarh) representing operational and temporal drivers identified from the steel-plant dataset.

## Results and Analysis

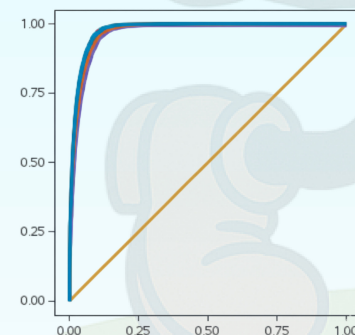


| Effect                              | Point Estimate | 95% Wald Confidence |
|-------------------------------------|----------------|---------------------|
| Load_Type Light_Load vs Medium_Load | 0.026          | 0.023 - 0.030       |
| Load_Type Max_Load vs Medium_Load   | 0.635          | 0.586 - 0.688       |
| Day_of_week Friday vs Wednesday     | 0.996          | 0.876 - 1.133       |
| Day_of_week Monday vs Wednesday     | 0.852          | 0.750 - 0.967       |
| Day_of_week Saturday vs Wednesday   | 2.578          | 2.198 - 3.024       |
| Day_of_week Sunday vs Wednesday     | 1.902          | 1.548 - 2.337       |
| Day_of_week Thursday vs Wednesday   | 0.91           | 0.801 - 1.033       |
| Day_of_week Tuesday vs Wednesday    | 0.915          | 0.805 - 1.041       |
| Usage_kWh                           | 1.024          | 1.023 - 1.026       |
| Month                               | 0.916          | 0.905 - 0.926       |
| Leading_CRP_kVarh                   | 3.237          | 3.067 - 3.418       |
| Day                                 | 1.012          | 1.007 - 1.016       |

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## Power Factor Efficiency

Leading reactive power was the most influential factor, with higher values linked to better power-factor efficiency. Load type also had a strong effect. Medium-load operations were most efficient, while both light-load and maximum-load periods were less likely to meet the 0.9 efficiency threshold. Time patterns showed that weekends, especially Saturdays and Sundays, had higher efficiency, while Mondays were slightly lower. Efficiency tended to decline over the months but improved slightly across days within each month. Overall, the results suggest that balanced operations and moderate reactive power contribute to more stable and efficient performance.



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| Load_Type Light_Load vs Medium_Load | 0.03           | 0.023 - 0.041       |
| Load_Type Max_Load vs Medium_Load   | 0.473          | 0.407 - 0.550       |
| Day_of_week Friday vs Wednesday     | 0.949          | 0.754 - 1.194       |
| Day_of_week Monday vs Wednesday     | 0.656          | 0.521 - 0.826       |
| Day_of_week Saturday vs Wednesday   | 2.283          | 1.582 - 3.295       |
| Day_of_week Sunday vs Wednesday     | 1.466          | 0.724 - 2.968       |
| Day_of_week Thursday vs Wednesday   | 0.873          | 0.698 - 1.092       |
| Day_of_week Tuesday vs Wednesday    | 0.793          | 0.631 - 0.996       |
| Month                               | 0.82           | 0.803 - 0.837       |
| Day                                 | 0.975          | 0.966 - 0.983       |
| Leading_CRP_kVarh                   | 0.58           | 0.344 - 0.978       |
| Lagging_CRP_kVarh                   | 1.183          | 1.175 - 1.192       |

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## Peak Loads

Lagging reactive power was the strongest factor, where higher lagging kVarh increased the chance of high-load events. Load type also played an important role. Medium-load operations were most likely to experience peaks, while light-load periods almost never triggered them. Time patterns were also clear, with fewer high-load events occurring later in the year and slightly less across each month. Saturdays showed the highest chance of peaks, while Mondays and Tuesdays were lower. Higher leading reactive energy reduced the odds of high-load events, suggesting a balancing effect.

## Conclusion and Contributions

### High Power Load (Peak) Findings:

- Peaks are not random; strongly linked to increases in Lagging\_CRP\_kVarh (lagging reactive power).
- Medium\_Load state shows highest association with peak events, not Maximum\_Load suggesting a unique process active during this state.
- Time patterns: Peaks occur more often on Saturdays and earlier in the year, indicating specific time windows for intervention.

### Power Factor Efficiency Findings:

- Efficiency (Lagging CPF > 0.90) peaks during Medium\_Load and weekends.
- Light\_Load periods show the lowest efficiency, aligning with power-system theory.
- Leading\_CRP\_kVarh is the strongest predictor of efficiency, showing capacitive correction is effective but insufficient at Light\_Load levels.

### Actionable Insights:

- Provides "why and when" operational inefficiencies occur, complementing predictive models that only show "what."
- For peak management: focus investigation on Medium\_Load operations and Saturday processes instead of generic power saving measures.
- For efficiency management: prioritize improvements during Light\_Load periods to prevent utility penalties.

## Problem Statement

The steel plant experiences ongoing fluctuations in electricity use and power quality. Two major challenges have been identified: **peak demand periods** and low **power factor** episodes. During peak demand, electrical loads rise well above normal levels, increasing costs through higher demand charges and placing additional strain on equipment and the grid. Low power factor events occur when inductive machinery causes current to lag voltage, reducing the efficiency of power use. This leads to higher current draw, greater transmission losses, and potential utility penalties. Together, these issues emphasize the importance of improving both load management and power factor efficiency to reduce costs and maintain reliable operations.

## Data Preparation

### Data Source

- 35,040 fifteen-minute records from a small steel plant in Gwangyang, South Korea.
- Includes real energy use (kWh), lagging/leading reactive energy, lagging/leading PF, CO<sub>2</sub>, and time features (NSM, weekday/weekend, day-of-week, and a three-level load type)

### Data Preparation

- Renamed column headers to load into SAS Viya
- Checked for duplicates and any missing values
- Split date into additional columns such as day, month and time

### Data Exploration

- Correlation analysis to identify pairwise numeric variables that were highly correlated
- Removed CO<sub>2</sub>\_tCO<sub>2</sub>, Lagging\_CRP\_kVarh and Leading\_CPF for binary logistic regression
- From Literature Review, 0.9 Lagging\_CPF and 95<sup>th</sup> percentile power usage were identified for power efficiency and high load usage respectively
- Created 2 flags based on the efficiency and load usage, and checked for complete separation

is\_inefficient

is\_high\_load

## 2 Explanatory Models

## Limitations and Future Work

The models are statistically strong but rely on high-level categories like Load\_Type. Future work should focus on using more detailed machine-level data, such as power usage by production floor and individual machines, to pinpoint which assets cause peaks or low power factor.

Adding operational context, like the number of workers on site or specific production schedules, could also help explain patterns such as the Saturday effect. Overall, this explanatory approach turned raw IoT data into a clear story of how the plant operates and provided a solid foundation for more targeted, data-driven efficiency improvements.